WEEK 3: PROJECT PLAN V1

Inclusive Reading Assistant

HAI Principle: User Control and Agency



Empowering university students to navigate dense academic texts through explicit, user-directed AI adaptations.

PROJECT OVERVIEW

The Problem

University students face severe cognitive overload when reading dense, jargon-heavy academic papers. Traditional reading takes excessive time, leading to fatigue.

The Solution

A web-based inclusive reading assistant operating strictly on a **"User Request Only"** basis, ensuring high agency before any AI intervention.

CRITICAL RULE: No adaptation happens without the user's explicit choice.



1. Simplify

Rewrites complex paragraphs into clearer English. *Aims to preserve meaning, but AI errors are possible. Original text is always restorable.*



2. Summarize

Condenses long sections into core arguments. Users strictly control the length and depth of the generated output.



3. Explain

Defines highlighted jargon or methodologies contextually without altering the surrounding source text.

USER PERSONA 1: AMIRA HASSAN

AH

Amira Hassan

First-year university student • Novice AI User • English as a Second Language

Background & Context

- First-year undergraduate balancing general education.
- Reads primarily on a smartphone.
- Relies on mobile networks with sometimes slow internet.
- Hesitant about AI; worries about incorrect information.

Accessibility & Needs

- **Cognitive-load sensitivity:** Easily overwhelmed.
- Requires simpler paragraph structures to parse meaning.
- Needs a seamless way to compare AI text with the original and restore it instantly.

USER PERSONA 2: OMAR KHALED

OK

Omar Khaled

Third-year university student • Typical Tech Level • Novice AI User

Background & Context

- Junior majoring in a social science discipline.
- Reads primarily on a laptop.
- Access to stable, fast internet on campus.
- Has not integrated AI tools into his daily study routine.

Accessibility & Needs

- Needs reliable summaries to triage papers efficiently.
- Requires on-demand explanations for theoretical frameworks.
- **Frustration:** Deeply worried about missing crucial nuances if an AI auto-summarizes.

USER PERSONA 3: MAYA CHEN

MC

Maya Chen

Graduate Student • Advanced Tech Level • Experienced AI User

Background & Context

- Master's student conducting primary research.
- Reads on a laptop with a dual-monitor setup.
- Stable, high-speed internet.
- Understands prompting and hallucination risks deeply.

Accessibility & Needs

- Demands granular control over AI parameters (length, level).
- Requires specific technical terminology to remain intact.
- **Frustration:** Consumer tools strip out statistical nuance and act as a "black box".

JOURNEY MAP: AMIRA (SIMPLIFY FOCUS)

1. Start

Pain: High cognitive load.
Design: Clean UI. Text loads normally. AI waits for explicit input.

2. Ask

Pain: Trust & latency.
Design: Tooltip explains action. Loading indicator assures progress.

3. Understand

Pain: Losing connection to original text.
Design: Output appears side-by-side. Source accessible.

4. Fix

Pain: Fear of permanent alteration.
Design: Clear "Restore Original" button ensures high reversibility.

5. Finish

Success: User completed task independently. Session resets cleanly.

JOURNEY MAP: OMAR (SUMMARIZE FOCUS)

Slide 7

1. Start

Pain: Wasting time on irrelevant papers.
Design: Select specific sections to summarize, not forced whole-document.

2. Ask

Pain: Lack of input context.
Design: Preference toggle before running: "Brief" vs "Detailed".

3. Understand

Pain: Summary missing specific details.
Design: Bullet points include anchor links to exact source paragraphs.

4. Fix

Pain: AI limitation (missing facts).
Design: Anchor links let user jump to source to verify missing details.

5. Finish

Success: Confident triage. Export option saves summary alongside document.

JOURNEY MAP: MAYA (EXPLAIN FOCUS)

1. Start

Pain: Context switching to Google.
Design: User highlights specific word and clicks "Explain" inline.

2. Ask

Pain: Generic outputs unhelpful.
Design: System automatically passes surrounding paragraph as context.

3. Understand

Pain: Tone mismatch.
Design: UI provides refinement chips: [More technical] [Give example].

4. Fix

Pain: Trapped by first-pass output.
Design: System regenerates instantly based on explicit chip feedback.

5. Finish

Success: Precise knowledge gained. Clicks away, resumes seamless reading.

MEASURABLE PROJECT OBJECTIVES

NOTE: The following are PROPOSED TARGETS for final evaluation, not achieved results.

Explicit User Choice

Proposed Target: 100% of adaptations are initiated by an explicit user choice. Zero background modifications.

Meaning Preservation

Proposed Target: At least 90% of evaluated outputs preserve the original text's key meaning without hallucinations.

Reversibility & View Control

Proposed Target: Users can view the original text and seamlessly restore it after adaptation in 100% of workflows.

EVALUATION METRICS

Metric	Category	Measurement Method	Proposed Target
Groundedness / Meaning Preservation	AI QUALITY	Manual assessment comparing generated output to source text key claims.	**≥ 90%**
Citation Accuracy	AI QUALITY	Manually verify generated summaries correctly map back to source citations.	**≥ 90%**
Task Success Rate	HUMAN-CENTERED	Percentage of users successfully completing intended reading task.	**≥ 85%**
User Control Compliance	PRINCIPLE-LINKED	Code audit to ensure adaptations are initiated ONLY by explicit user choice.	**100%**
Error Recovery Rate	USABILITY	Percentage of simulated error states where users use a recovery path.	**≥ 90%**

HAI PRINCIPLE LADDER & OBJECTIVES

Slide 11

Level	Concrete Feature	Supports Agency By	Measurable Objective	Proposed Target	Method	Test Procedure	Status
1. INFORM	Tooltips explicit defining Simplify vs. Summarize.	Ensuring users understand AI actions.	Users understand functions before choosing.	≥85% comprehension	Pre-task questionnaire.	Ask to explain tool diffs.	PROPOSED
2. EXPLAIN	Visual disclaimers on adapted text.	Setting expectations of AI fallibility.	Users recognize result may contain errors.	≥90% recognition	Post-task interview.	Ask who wrote the text.	PROPOSED
3. CONTROL	Manual button click. No auto-run.	Preventing unwanted background alterations.	100% adaptations initiated by explicit choice.	100% compliance	Code audit.	Verify no auto-load.	PROPOSED
4. FEEDBACK	"Restore Original" & refinement chips.	Allowing seamless recovery from unwanted results.	Users can successfully restore original text.	≥90% success rate	Task completion tracking.	Instruct user to undo.	PROPOSED

PROPOSED TOOLS & ARCHITECTURE

NOTE: SELECTIONS REMAIN PROPOSED AND PENDING CODEBASE IMPLEMENTATION

Architecture Details (Proposed)

- **LLM (Proposed):** Existing foundational LLM accessed strictly via API. *No local training from scratch.*
- **Prompts (Proposed):** Three isolated, distinct system prompts for Simplify, Summarize, and Explain to prevent cross-contamination.
- **Security (Pending Check):** API keys stored securely in environment variables, never committed to Git.
- **Frontend/Backend:** Web-based React/Vue frontend routed through a Node/Python server.

Data Strategy (Proposed)

- **Sample Texts:** Proposed to use a small subset (~30 papers) of public open-access academic literature.
- **Compliance (Pending Check):** All texts subject to strict licensing and permitted-use verification before integration.
- **Grounding:** System will pass specific paragraph chunks contextually to the LLM to ground the Explain function.

DELIVERABLES & NEXT STEPS

Completed: Week 3 Plan

- Defined three distinct fictional personas.
- Mapped three end-to-end user journeys highlighting user control.
- Established measurable **proposed** objectives.
- Developed the HAI Principle Ladder v1.
- Documented Bias-Checks and Data Source proposals.

Next Milestone: Week 4

- Design the detailed conversational workflow.
- Map out comprehensive system architecture and APIs.
- Develop explicit error handling and undo paths.
- Evaluate UI against Shneiderman's Eight Golden Rules.
- **Technical Goal:** Retrieve the first real model response successfully from the backend.

APPENDIX A: BIAS-CHECK NOTES

Area Reviewed	Potential Bias / Exclusion	Evidence / Observation	Change Made / Recommended	Status
Names & Culture	Inferring tech ability from ethnic background.	Varied names used (Hassan, Khaled, Chen) with distinct backgrounds.	Explicitly mapped tech proficiency to experience level, not background.	REVIEWED
Accessibility & Devices	Designing only for high-end laptops, excluding low-income users.	Amira relies on a smartphone with sometimes slow internet.	Added requirement for latency tracking and lightweight UI design.	REVIEWED
Language vs. Cognition	Assuming ESL learners need "dumbed down" intellectual content.	Amira has cognitive-load sensitivity and ESL needs.	Simplify vocabulary while strictly preserving academic complexity.	PROPOSED RISK
User Preferences	Assuming all novice AI users want identical assistance.	Omar worries about missing info; Amira worries about overload.	Included preference toggles (e.g., length control).	PROPOSED CHECK

APPENDIX B: DATA SOURCE DESCRIPTION

STATUS: PROPOSED — PENDING TEAM VERIFICATION

1. What the Data Contains & Source

Proposed Source: Semantic Scholar Open Research Corpus (S2ORC). Contains rich metadata and structured full-text for open-access academic papers across disciplines.

2. Dataset Size & License

Full Size: 81.1M total papers (8.1M with parsed full-text).
Our Target Subset: A tailored set of ~30 open-access texts for evaluation.
License: Metadata under ODC-BY; full text varies by publisher. *(Status: Pending Verification of permitted use for target subset).*

3. Why It Fits the Project

Provides authentic, dense academic English matching our target domain. Perfect for evaluating Simplify and Summarize against real jargon.

4. Limitations & Bias

- May over-represent Western-centric or English-first publishing patterns.
- Risk of outdated methodologies.
- **Privacy:** Public papers contain author names; must ensure no private PII is inadvertently processed.

APPENDIX C: PROMPT LOG (WEEK 3)

Documenting prompts used for concept generation. **Team must retrieve missing historical exact prompts.**

ID	Date	Exact Prompt	Tool / Model	Actual Output / Summary	Team Change / Reason	Status
001	Wk 2	*Exact prompt unavailable — team must retrieve from logs.*	Claude	Drafted "ClearPath Reader" interaction concept.	Selected ClearPath over AdaptRead. Aligned better with User Agency.	MISSING DATA
002	Wk 3	*[Team: Insert exact prompt used for Persona generation]*	*[Pending]*	*[Team: Insert raw output summary]*	*[Team: Insert changes and reasoning]*	MISSING DATA
003	Future	*[Blank Template for Future Feature Prompts]*	*[Blank]*	*[Blank]*	*[Blank]*	TEMPLATE

REQUIREMENTS COMPLIANCE CHECKLIST

Requirement	Status
Three distinct fictional personas & Journey Maps	COMPLETE
Measurable objectives with numeric targets	COMPLETE
HAI Principle Ladder linked to tests	COMPLETE
Bias-check notes (Review, findings, changes)	COMPLETE
Data-source description (Size, license, subset fit)	PENDING LICENSE CHECK
Architecture, Models, Backend, and Grounding logic	PENDING CODEBASE CHECK
Exact prompt-log entries, tools, outputs	MISSING / NEEDS RETRIEVAL